

Problem Set 12 (40 pts)

Due at 11:59pm

Friday, December 5

Exercises are originally from <https://aimacode.github.io/aima-exercises/> after some fixing, rewriting, and adaptations.

1. **(6 pts)** (Exercise 14.22(a)) This exercise explores the stationary distribution for Gibbs sampling methods. A convex composition of two transition kernels k_1 and k_2 is a transition probability distribution that first chooses one of k_1 and k_2 with probabilities α and $1 - \alpha$, for some $0 \leq \alpha \leq 1$, respectively, and then applies the chosen kernel. Prove that if k_1 and k_2 are in detailed balance with a stationary distribution π , then their convex composition is also in detailed balance with π .
2. **(9 pts)** (Exercise 15.2(a)) In this exercise, we examine what happens to the probabilities in the umbrella world in the limit of long time sequences. Suppose we observe an unending sequence of days on which the umbrella appears. The probability of rain on the current day increases monotonically toward a fixed point. Calculate this fixed point using the filtering formula in the notes titled “temporal models” (see p. 12 in .pptx and p. 74 in .pdf) on November 19.
3. **(25)** (Parts of Exercises 16 – 17) A professor wants to know if students are getting enough sleep. Each day, the professor observes whether the students sleep in class, and whether they have red eyes. The professor has the following domain theory:
 - The prior probability of getting enough sleep, with no observations, is 0.7.
 - The probability of getting enough sleep on night t is 0.8 given that the student got enough sleep the previous night, and 0.3 if not.
 - The probability of having red eyes is 0.2 if the student got enough sleep, and 0.7 if not.
 - The probability of sleeping in class is 0.1 if the student got enough sleep, and 0.3 if not.

Introduce three boolean variables: S_t , whether the students get enough sleep; R_t , whether they have red eyes in class; and C_t , whether the students sleep in class.

(a) (10 pts) Formulate this information as a dynamic Bayesian network that the professor could use to filter from a sequence of observations. You may draw a portion of the Bayes net, e.g., from time $t - 1$ through time $t + 1$, with necessary conditional probability tables, or you may describe its structure and list all the necessary conditional probabilities.

(b) (15 pts) You are given the following evidence values:

- e_1 = not red eyes, not sleeping in class
- e_2 = red eyes, not sleeping in class
- e_3 = red eyes, sleeping in class

Use the same filtering formula to compute the probabilities $P(S_t \mid e_{1:t})$ for $t = 1, 2, 3$.